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DETAIL SPECIFICATION

IMPULSE CARTRIDGE, CCU-72/A

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the CCU-72/A impulse cartridge, referred to herein as "cartridge."

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

MIL-C-83124 - Cartridge Actuated Devices/Propellant Actuated Devices General Design Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-464 - Electromagnetic Environmental Effects Requirements for Systems

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

841AS450 - Test Tip, Shielded Mild Detonating Cord

850AS132 - Cartridge, Impulse CCU-72/A

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandsStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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850AS834 - Test Fixture Impulse Cartridge CCU-72/A

850AS836 - Vibration Fixture Impulse Cartridge CCU-72/A

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E748 - Standard Guide for Thermal Neutron Radiography of Materials

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AS5604 - Compass; Magnetic, Pilot's Standby

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design and construction. The cartridge shall be manufactured in accordance with Drawing 850AS132.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform to the standard established at technical service release as specified (see 6.2). Any divergence shall require technical evaluation and potential design certification testing.

3.4 Primary components. All explosive materials used in the manufacture of this item are considered primary components. Only primary components from a single lot shall be used in a cartridge lot.

3.4.1 Primers. Primers used in the cartridge shall be of the type originally approved and from lots manufactured within 24 months of cartridge loading date or from primer lots recertified within 12 months of cartridge loading date.

3.5 Cartridge requirements.

3.5.1 Visual requirements. All cartridges shall have adequate and legible descriptive markings, serial numbers, and color coding. Cartridges shall not contain any dents, deep scratches, or other defects that may prevent proper installation or function.

3.5.2 Leakage. Cartridges shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second (cc/sec) of air when tested in accordance with 4.5.3.

3.5.3 Radiographic requirement. Cartridges shall be free of imperfections in assembly when examined radiographically in accordance with 4.5.4.

3.5.4 Demagnetization. Cartridges shall not deflect an approved pilot's standby compass more than three degrees in either direction when passed in front of the compass. The cartridges shall be tested in accordance with 4.5.5.

3.5.5 Electrical bonding resistance. Cartridges shall not exhibit an electrical resistance in excess of 0.0025 ohm when tested in accordance with 4.5.6.

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3.6 Environmental requirements.

3.6.1 Vibration. The cartridge shall not fire or suffer damage which could adversely affect the form, fit, or function of the cartridge when subjected to the vibration environment in accordance with the requirements of 4.5.7. After being subjected to the vibration environment, the cartridges shall meet the requirements of 4.5.10 when ballistically tested.

3.6.2 Temperature/humidity cycling. The cartridge shall not fire or suffer damage which could adversely affect the form, fit, or function of the cartridge when subjected to the temperature/humidity environment of MIL-D-21625 and tested in accordance with 4.5.8. After being subjected to the temperature/humidity environment, the cartridges shall meet the requirements of 4.5.10 when ballistically tested.

3.6.3 Shock. The cartridge shall not fire or suffer damage which could adversely affect the form, fit, or function of the cartridge when subjected to the shock environments of MIL-D-21625 when tested in accordance with 4.5.9. After being subjected to the shock environment, the cartridges shall meet the requirements of 4.5.10 when ballistically tested.

3.7 Ballistic requirements. Cartridges from each first article and production lot shall meet the requirements of 3.7.1 through 3.7.4 when subjected to the test prescribed in 4.5.10.

3.7.1 Maximum pressure. The upper (U) and lower (L) limits for maximum pressure (psig) when fired over the temperature range from −65 to +200 °F shall be U = 4000 psig and L = 2600 psig, respectively (see 4.5.10.1.1).

3.7.2 Ignition delay. The limit for ignition delay shall not be greater than 20 milliseconds (see 4.5.10.1.2).

3.7.3 Time to maximum pressure. The time to maximum pressure will be equal to or less than 60 milliseconds when fired across the temperature range (see 4.5.10.1.3).

3.7.4 Misfire. There shall be no misfires (see 4.5.10.1.4).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article sample. Unless otherwise specified (see 6.2), an acceptable first article sample of 72 cartridges shall be prepared using the same procedures and methods proposed for the production lot. Sixty-six of these cartridges shall be expended in the tests listed in table I and six shall be retained for investigative purposes. A first article sample acceptable for functional and environmental testing shall be defined as a sample that has met the requirements of 3.2 through 3.5 of this specification. First article testing is for the purpose of determining that the contractor, prior to starting production, is capable of producing items that comply with the technical requirements as specified in the contract. As specified (see 6.2), a first article inspection is required if two or more years have passed since the item was manufactured, after major turnover in personnel, following major changes to the production facility, or following transfer of production to a new facility or site. Failure of any cartridge to comply with the requirements of table I shall be cause for rejection of the first article represented.

TABLE I. First article inspection.

Test/inspections	Requirement paragraph	Test paragraph	Sample groups				
			A	B	C	D	E
Visual	3.5.1	4.5.2	12	18	12	12	12
Leak	3.5.2	4.5.3	12	18	12	12	12
Radiographic	3.5.3	4.5.4	12	18	12	12	12

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TABLE I. First article inspection - Continued.

Vibration	3.6.1	4.5.7	12				
Temperature/humidity	3.6.2	4.5.8		18			
Shock	3.6.3	4.5.9			12		
Ballistic (−65±5 °F)	3.7	4.5.10	6	9	6	12	
Ballistic (200±5 °F)	3.7	4.5.10	6	9	6		12

4.3 Production inspection. All cartridges shall be inspected and screened to the examinations and tests listed in table II. Any cartridge which fails to meet the requirements of table II shall be rejected and removed from the lot.

TABLE II. Production inspection.

Item	Test	Requirement paragraph	Test paragraph
Primary components	Single lot	3.4	4.5.1
Cartridge	Visual	3.5.1	4.5.2
Cartridge	Leak	3.5.2	4.5.3
Cartridge	Radiographic	3.5.3	4.5.4
Cartridge	Demagnetization	3.5.4	4.5.5
Cartridge	Electrical bonding resistance	3.5.5	4.5.6

4.4 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests in table III. Failure of any sample to comply with the requirements of table III shall be cause for rejection of the lot represented.

TABLE III. Lot acceptance inspection.

Test/inspections	Requirement paragraph	Test paragraph
Visual	3.5.1	4.5.2
Leak	3.5.2	4.5.3
Radiographic	3.5.3	4.5.4
Ballistic (-65±5 °F)	3.7	4.5.10
Ballistic (70±5 °F)	3.7	4.5.10
Ballistic (200±5 °F)	3.7	4.5.10

4.4.1 Sampling. A random sample of cartridges, including the samples for investigative purposes, shall be selected from the production lot in accordance with table IV. An additional three cartridges shall be provided with each test sample for investigative purposes.

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TABLE IV. Sample sizes.

Lot size	Normal sample size
2 to 8	2
9 to 15	3
16 to 25	5
26 to 50	8
51 to 90	13
91 to 150	20
151 to 280	32
281 to 500	50

4.5 Inspections and tests.

4.5.1 Primary components. Use of a single lot of primary components shall be verified by inspection of the materials used for each first article or production lot.

4.5.2 Visual inspection. All cartridges shall be examined, and all those with illegible descriptive markings or serial numbers, color coding, dents, deep scratches, or other defects that may prevent proper installation or function shall be rejected.

4.5.3 Leakage. All cartridges shall be leak tested in a dry gas leak tester. Cartridges that exhibit a leak rate in excess of 1×10^{-5} cc/sec of air at a pressure differential of 1±0.1 atmosphere shall be considered defective. If one or more cartridges of the ballistic sample leak, then a leak test of each cartridge in the production lot shall be required. Cartridges that exhibit a leak rate in excess of that specified above shall be rejected and destroyed.

4.5.4 Radiographic examination. All cartridges shall be identified with serial numbers prior to radiographic examination. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The cartridges shall be positioned to give the most revealing cross-section of the cartridge. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiographic plate. Defective cartridges shall be clearly identified on the radiographic plate and those defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test as specified (see 6.2).

4.5.4.1 X-ray examination. Each cartridge shall be x-rayed in accordance with ASTM E1742/E1742M to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for cartridge rejection.

4.5.4.2 N-ray examination. Each cartridge shall be n-rayed in accordance with ASTM E748 to confirm the presence of the propellant, non-metal parts, no foreign hydrogenous materials, and the proper assembly of pyrotechnic/non-metallic materials. Any observable imperfection shall be cause for rejection.

4.5.5 Demagnetization. Each cartridge shall be moved past a pilot's standby compass face (in accordance with SAE AS5604) by placing the cartridge 6 inches from the compass face, in a north-south horizontal position with the compass set in an east-west heading. The cartridge and the compass shall be placed at the same height from the floor. The cartridges shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved back past the compass face. This procedure shall be repeated for every 90 degrees of the cartridge.

4.5.6 Electrical bonding resistance. Each cartridge shall be tested using a four-point probe milliohm meter in accordance with MIL-STD-464. The electrical bonding resistance shall be measured between the thread of the SMDC inlet port and the thread at the outlet end of the cartridge.

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4.5.7 Vibration. The vibration test shall be performed on all 12 cartridges at the same time. The 12 cartridges shall be mounted in the vibration test fixture (in accordance with Drawing 850AS836) for the cartridge. The test specimens shall be exposed to the sinusoidal vibration environment shown in table V and on figure 1 with the frequency varying logarithmically from 5 to 2,000 hertz and returning to 5 hertz in approximately 20 minutes. The specimens shall be exposed to the environment for 1 hour in each of the three mutually perpendicular axes, at ambient, +200 °F, and -65 °F, for a total of 9 hours of testing.

TABLE V. Sinusoidal vibration environment.

Frequency (hertz)	Acceleration/amplitude
5 to 20	0.10 inch D.A. 1/
20 to 33	± 2 g's peak
33 to 74	0.036 inch D.A. 1/
74 to 2,000	± 10 g's peak
NOTE: 1/ D.A. = double amplitude	

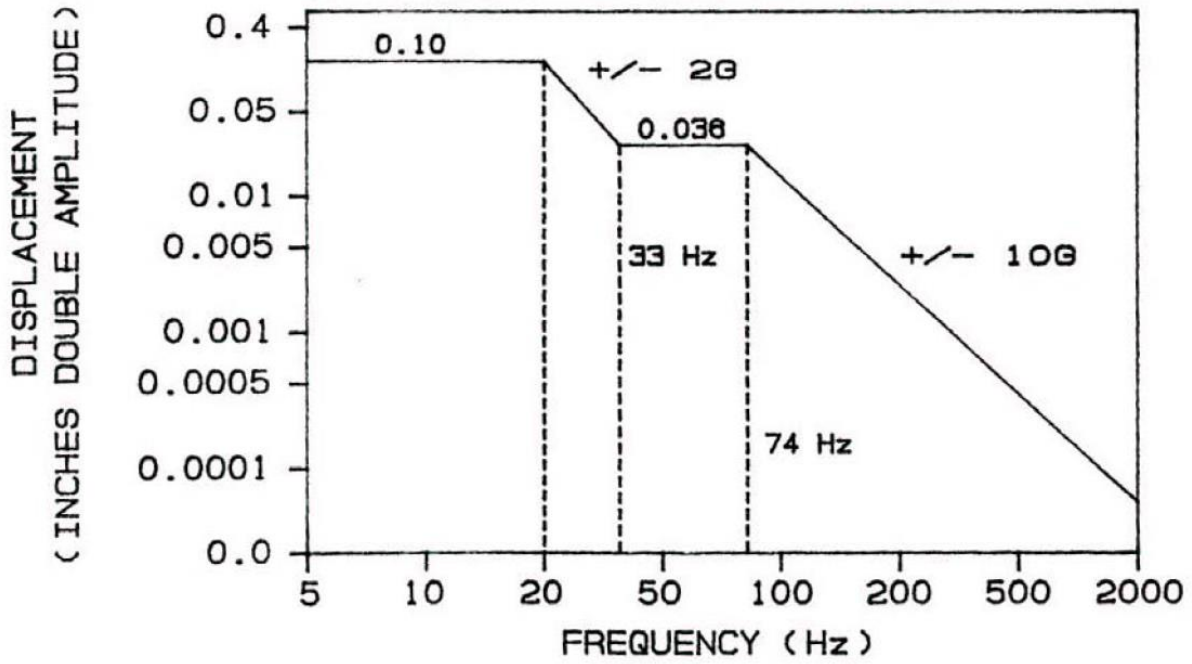


FIGURE 1. Non gunfire vibration.

4.5.8 Temperature/humidity cycling test. Test cartridges shall be subjected to temperature/humidity environments in accordance with MIL-C-83124, except that the altitude shall be 0 feet (14.7 pounds per square inch absolute). If any initiator fails to meet the requirements of 3.6.2, the first article sample shall be rejected.

4.5.9 Shock test. Test cartridges shall be subjected to shock tests in accordance with method 516, procedure I of MIL-STD-810 with the safety pin removed. The shock pulse wave shall be a terminal peak sawtooth. The peak amplitude shall be 20g and the duration shall be 11 milliseconds. If any initiator fails to meet the requirements of 3.6.3, the first article sample shall be rejected.

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4.5.10 Ballistic test. Cartridges shall be selected from each production lot in accordance with 4.4.1. The sample of cartridges shall be divided, as nearly as possible, into three equal groups. If the sample of cartridges is not equally divisible by three, the excess cartridge(s) shall be placed in the group that is to be fired at -65±5 °F. One group each shall be conditioned for a period of time neither less than 4 hours nor greater than 24 hours at -65±5 °F and +200±5 °F. The other group shall be conditioned for a period of time not less than 4 hours at +70±5 °F. The cartridges shall be fired in the closed bomb test fixture (in accordance with Drawing 850AS834), set up as shown on figure 2 below, within 3 minutes after removal from temperature conditioning. The cartridges shall be temperature conditioned with the SMDC test tip (in accordance with Drawing 841AS450) installed. If any cartridge is exposed to the temperature environment for greater than the time allowed or not fired within 3 minutes after removal, it shall be returned to a +70±5 °F environment for at least 4 hours and then re-exposed to the original environment for not less than 4 hours and not more than 24 hours before test firing. The re-exposure may be performed as many times as necessary to complete the required ballistic test. Instrumentation shall be provided to measure and record the cartridge pressure (psig) in the test chamber, ignition delay (milliseconds), and time to maximum pressure (milliseconds). The test chamber shall be thoroughly cleaned after each cartridge firing. The production lot shall be accepted if the sample fulfills the requirements of 4.5.10.1.1 through 4.5.10.1.4.

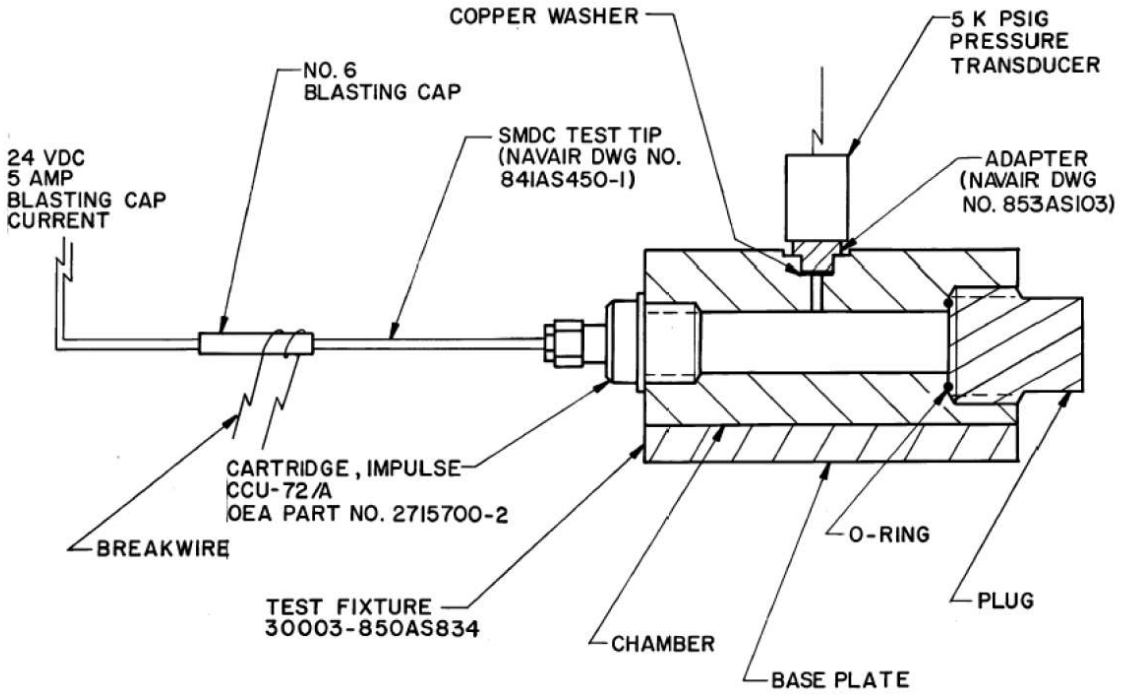


FIGURE 2. Test setup.

4.5.10.1 Acceptance criteria.

4.5.10.1.1 Maximum pressure. The maximum pressure shall be measured at the point that the pressure reaches its maximum in the test fixture. The lot shall be acceptable with respect to maximum pressure if the ballistic sample meets the requirements of 3.7.1.

4.5.10.1.2 Ignition delay. The ignition delay shall be measured from the actuation of the blasting cap to first indication of pressure. The lot shall be acceptable with respect to ignition delay if the ballistic sample meets the requirements of 3.7.2.

4.5.10.1.3 Time to maximum pressure. The time to maximum pressure shall be measured from the first indication of pressure to the point that the pressure reaches its maximum in the test fixture. The lot shall be acceptable with respect to time to maximum pressure if the ballistic sample meets the requirements of 3.7.3.

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4.5.10.1.4 Misfire. A cartridge that does not fire under the conditions specified in 4.5.10 shall be considered defective, and the lot considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.5.10.2 Retest. There shall be no retests.

4.5.10.3 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridge is initiated by one shielded mild detonating cord (SMDC) assembly to actuate the pilot's canopy unlatch thruster, which unlocks the canopy before canopy jettison in the F-18 aircraft. The assembly was designed for use in military aircraft and has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the source control drawing and this specification.

b. When first article inspection is required (see 3.1 and 4.2).

c. Materials required as established by the technical service release (see 3.3).

d. Production lot size, sample size, and test activity.

e. Direction for shipping radiographic plates of entire production lot along with ballistic sample to the activity conducting production lot acceptance test (see 4.5.4).

f. Items of data required for each first article and production lot.

g. That the safety requirements of DoD Manual 4145.26 are applicable (see 6.3).

6.3 Safety precautions. The safety precautions requirements of DoD Manual 4145.26, "Contractor's Safety Manual for Ammunition and Explosives," are applicable and should be specified in the contract as required by the Defense Federal Acquisitions Regulation (DFAR) 1-323 (see 6.2). When this specification is used as part of the description of work to be accomplished by a government activity, the safety precaution requirements of NAVSEA OP 5, "Ammunition and Explosives Safety Ashore," should be made applicable.

6.4 Explanation of terms. For the purpose of this document, NAVSUPINST 4030.28, Packaging of Materiel, when applicable should be used to clarify terms.

6.5 Subject term (key word) listing.

Canopy jettison
Ejection systems, aircrew
SMDC
F-18

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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